

Tribulus terrestris can suppress the adverse effect of toluene on bovine and equine ovarian granulosa cells

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Abstract

Influence of oil-related product toluene and herbal remedy puncturevine *Tribulus terrestris* L. (TT) on female reproduction is known. Yet, mechanisms of their action on ovaries in different species and potential protective effect of TT against adverse toluene action remain to be established. We studied the effect of toluene, TT, and their combination on ovarian granulosa cells from two mammalian species (cows and horses). Viability, markers of proliferation (PCNA) and apoptosis (bax), steroid hormones, IGF-I, oxytocin, and prostaglandin F (PGF) release were analyzed by trypan blue exclusion test, quantitative immunocytochemistry, and EIA/ELISA. Toluene suppressed all analyzed parameters. In both species, TT stimulated proliferation and reduced progesterone, oxytocin, and PGF. In horses, TT inhibited testosterone and IGF-I. In both species, TT supported toluene effect on viability, steroids, IGF-I, and PGF, and inverted its action on apoptosis. In cows, TT promoted toluene effect on proliferation. In horses, TT supported toluene effect on oxytocin but suppressed its influence on proliferation. In both species, toluene induced inhibitory action of TT on viability, steroids, IGF-I, and PGF, and prevented its stimulatory action on proliferation. In cows, toluene supported inhibitory action of TT on oxytocin and prevented its stimulatory action on apoptosis. In horses, toluene induced stimulatory effect of TT on apoptosis. Our results indicate potential toxic toluene effect on farm animal ovaries, applicability of TT as a biostimulator of farm animal reproduction and as a protector against the adverse influence of toluene on female reproduction.

KEYWORDS

bovine ovary, equine ovary, ovarian hormones, toluene, *Tribulus terrestris* L.

1 | INTRODUCTION

Livestock species, including cows and horses, are important to humans from economic, social, and cultural points of view. Cows and horses can also be used as models for biomedical research. The action of environmental pollutants and the action of herbal remedies in the reproductive processes of cows and horses have been insufficiently studied. Investigation of their influences could help improve fertility and eliminate the risks linked to cow and horse reproduction.

One of the most common hazardous sources of environmental contamination is oil-related products, including toluene. Aromatic hydrocarbons are especially dangerous because of their numerous sources of contamination in the environment (e.g., automobile exhaust, evaporation of toluene-containing products from their production, storage, and waste, as well as from preparation of detergents and many other chemicals and pharmaceuticals) (WHO, 2005).

The adverse action of toluene on female reproductive processes at various regulatory levels has been well documented. Epidemiological studies have shown that female workers' exposure

to toluene results in a decrease in fertility (Sallmén et al., 2008), pre-ovulatory blood gonadotropin (Chen et al., 2001) and prostaglandin (PG) levels (Reutman et al., 2002), and an increased spontaneous abortion rate (Ng et al., 1992). The ability of inhaled toluene to increase maternal and prenatal morbidity and embryonic growth retardation has been demonstrated in rats (Roberts et al., 2003, 2007), mice (Donald et al., 1991), and cows (Waldner, 2008). Furthermore, toluene causes degenerative histomorphological changes in mouse ovarian tissue (Kareem et al., 2014). Furthermore, in vitro studies have demonstrated that the harmful influence of toluene on reproduction can be linked to its ability: (a) to alter the release of progesterone and estradiol (Sirotkin, Záhoranská, et al., 2021), progesterone, oxytocin, and prostaglandin F (PGF) in pigs (Sirotkin, Kadası, et al., 2020) and of progesterone, IGF-I, and PGF in horses (Tarko et al., 2020); (b) to reduce proliferation of porcine (Sirotkin, Záhoranská, et al., 2021), bovine (Sirotkin, Makarevich, et al., 2020), and equine (Tarko et al., 2020) ovarian cells; (c) to promote apoptosis of porcine ovarian cells (Tarko et al., 2020); and (d) to induce degenerative changes in antral ovarian follicles of rats (Tap et al., 1996). Despite the demonstration of the negative influence of toluene on animal and human female reproduction, it remains to be elucidated whether toluene can affect: (a) cell viability, testosterone, oxytocin, and PGF release by bovine ovarian cells; and (b) testosterone output by equine ovarian cells.

The herbal remedy puncturevine *Tribulus terrestris* L. (TT) contains numerous bioactive phytochemicals, such as saponins, phytosterols, tannins, flavonoids, glycoside alkaloids, and acids (Chhatre et al., 2014). Because of the plant's active substances, TT exhibits antioxidant, antibacterial, aphrodisiac, astringent, analgesic, anthelmintic, anti-inflammatory, anti-spasmodic, neuroprotective, hepatoprotective, and anticancer properties (Chhatre et al., 2014) and is considered a promising preventative option or treatment for metabolic, cardiovascular, immune, and neurodegenerative disorders (Jameel et al., 2012). Recent studies have shown that this plant can influence female reproduction and improve dehydroepiandrosterone levels in women (Gama et al., 2014). *Tribulus terrestris* L. (TT) prevents the development of ovarian cysts and polycystic ovarian syndromes in rats (Dehghan et al., 2012; Saiyed et al., 2016; Sandeep et al., 2015) and women (Arentz et al., 2014). Rabbits fed TT extract exhibited differences in the expression of oocyte-derived factors, which are presumed to participate in controlling follicular development and ovulation (Abadjieva & Kistanova, 2016). Oral treatment with an extract of TT increased the number of growing follicles (Adaay & Mosa, 2012) and reduced serum testosterone in mice (Bertozzo & Moreira, 2019). *Tribulus terrestris* L. (TT) can promote accumulation of proliferation marker (proliferating cell nuclear antigen, PCNA) and apoptotic marker (bcl-2-associated X protein, bax) in porcine ovarian cells (Sirotkin, Alexa, & Harrath, 2020). On the other hand, to our knowledge, the effect of TT on ovarian cell function in cows and horses (viability, proliferation, apoptosis, steroid, and peptide hormone secretion) remains unknown.

From a theoretical and practical viewpoint, it would be appropriate to find strategies for preventing and neutralizing the potential

adverse effects of toluene on the female reproductive system. Herbal remedies and functional foods are gaining great attention given their possible beneficial and therapeutic effects (Jameel et al., 2012). For example, the plant polyphenols apigenin, daidzein, and rutin, and the medicinal plant fennel can reportedly suppress toluene influence on cell viability, proliferation, apoptosis, and the release of estradiol, IGF-I, and PGF in porcine (Sirotkin, Záhoranská, et al., 2021) and equine (Tarko et al., 2020) ovarian cells. Furthermore, TT is able to provide effective protection against the toxic effects of cadmium (Lakshmi et al., 2012), lead acetate (Eagappan et al., 2015), and nifedipine (Aslam, 2018) in rats and cisplatin in mice (Keshtmand et al., 2014; Raoofi et al., 2015). Yet, the ability of TT to prevent or cure the adverse effects of toluene on reproductive processes has not yet been examined.

Our experiments aimed to address the following questions:

1. Does toluene, when added alone, directly affect basal bovine or equine ovarian granulosa cell functions [viability, proliferation (accumulation of PCNA), apoptosis (accumulation of bax), or the release of progesterone, testosterone, IGF-I, oxytocin, or PGF]?
2. Does TT, when added alone, affect these basic bovine and equine granulosa cell functions?
3. Does TT, when added in combination with toluene, modify or prevent the action of toluene on these cells? If such ability of TT were to be demonstrated, it would suggest the potential applicability of TT, or products containing TT, as natural protectors against the negative effect of toluene on reproductive system?
4. Are there species differences in the response of cultured bovine and equine ovarian granulosa cells to toluene, TT, or their combination?

2 | MATERIALS AND METHODS

2.1 | Preparation, processing, and culture of ovarian granulosa cells

In the first series of experiments, a subset of 16 ovaries from Holstein breed cows (3–5 years of age) at the follicular stage of the estrous cycle were obtained from the slaughterhouse of the Research Institute of Animal Production in Nitra and from the slaughterhouse in Stara Myjava (Slovakia). In the second series of experiments, a subset of 16 ovaries from horses (4–13 years of age) at the follicular stage of the estrous cycle (determined by visual inspection of the ovaries) were obtained from the slaughterhouse in Zakłady Mięsne Józan (Poland).

Ovarian granulosa cells were isolated from ovarian bovine (3–5.5 mm diameter) and equine (5–7 mm diameter) follicles without visible signs of atresia (weak vascularization, thin follicular wall, pale follicular fluid) by aspiration with a syringe. After aspiration and cell isolation by centrifugation for 10 min at 200 g, the granulosa cells were washed in sterile DMEM/F12 1:1 medium

(BioWhittakerTM; Lonza) and resuspended in the same medium supplemented with 10% fetal calf serum (BioWhittakerTM) and 1% antibiotic-antimycotic solution (Sigma-Aldrich). Cells were counted using a Buerker chamber, and their concentration was adjusted to the required volume (10^6 cells/ml medium). The cell suspension was distributed into 24-well culture plates (NuncTM; 1 ml suspension/well) for enzyme immunoassay (EIA)/enzyme-linked immunosorbent assay (ELISA) or onto 16-well chamber slides (Nunc Inc. International; 200 μ l per well) for immunocytochemistry and incubated at 37.5°C in 5% CO₂ until a 75% confluent monolayer was formed (3–4 days). After replacing the medium, the experimental cells were cultured in the presence of 99.9% toluene (AppliChem GmbH) at concentrations of 0, 1, 10, and 100 ng/ml or in the presence of a 95% extract of TT leaves (Changsha Sunfull Bio-Tech. Co.) at a concentration of 10 μ g/ml, alone, or in combination with toluene (1, 10, and 100 ng/ml). These doses correspond to the dose of toluene that is typically found in the environment (Bolden et al., 2015) and to the effective doses of toluene (Sirotkin, Záhoranská, et al., 2021; Tarko et al., 2020) and TT (Bertozzo & Moreira, 2019) used in previous in vitro experiments. Toluene (99.9%) was dissolved in culture medium, while 1 mg of dry powder of TT leaf extract was first suspended in dimethyl sulfoxide (DMSO) (AppliChem GmbH) and thereafter dissolved in culture medium up to a desirable concentration. Control groups contained ovarian cells cultured in incubation medium without either toluene or TT, while the blank control consisted of the cell-free medium. After 2 days of incubation, the culture medium from the 24-well culture plates was removed via syringe and stored at –18°C until analysis by EIA/ELISA. On the chamber slides, 4% paraformaldehyde in phosphate-buffered saline (PBS; Sigma-Aldrich) for 10 min was used to fix a monolayer of cells, which were then stored at +4°C until immunocytochemistry analysis. Cell concentration and viability were determined by trypan blue exclusion test and haemocytometer.

2.2 | Trypan blue exclusion test

The effect on cell viability of toluene, TT extract, or their combination was detected by the trypan blue exclusion test (0.4%), as described previously (Strober, 2015). Briefly, after incubation of the granulosa cells, the medium was removed from the culture plates. Subsequently, the cell monolayer was subjected to trypan blue staining (Sigma-Aldrich) for 15 min. Following the removal of this dye, plates were washed twice with a physiological solution and subjected to microscopic inspection (magnification: 400 $\times$). The data from trypan blue exclusion test are reported as the means of values obtained in three separate experiments performed on separate days with different groups of granulosa cells, each obtained from at least five ovaries. Each treatment group was represented by four wells, and the ratio of dead (stained) cells to the total cell count was calculated from at least 100 cells per well. Significant differences between the groups were determined using the Shapiro–Wilk normality and Student's *t* test and two-way ANOVA followed by Tukey's

test with SigmaPlot 11.0 (Systat Software GmbH). Differences were compared for statistical significance at *p* levels less than .05 (*p* < .05).

2.3 | Immunocytochemical analysis of proliferation and apoptosis markers

The presence of markers of proliferation (PCNA) and cytoplasmic apoptosis (bax) was detected by immunocytochemistry (Gaumer et al., 2000; Osborn & Brandfass, 2006). After washing and fixation, the cells were incubated in a blocking solution (1% goat serum [Santa Cruz Biotechnology Inc.] in PBS) at room temperature for 1 h to block non-specific antiserum binding. Thereafter, the cells were incubated in primary mouse monoclonal anti-PCNA or anti-bax (dilution 1:500 in PBS; Santa Cruz Biotechnology Inc. for both antisera) for 1 h at room temperature. For the detection of binding sites of the primary antibody, the cells were incubated in a secondary swine antibody against mouse IgG (dilution 1:1000; Santa Cruz Biotechnology Inc.) labelled with horseradish peroxidase (Servac) for 1 h. Positive signals were visualized by staining with 3,3'-diaminobenzidine (DAB) substrate (Roche Diagnostics GmbH) for 1 h. Next, the cells on the chamber slides were washed in PBS and covered with a drop of Glycergel mounting medium (DAKO) before a coverslip was placed onto each microslide. Using light microscopy, DAB-peroxide brown staining was performed to determine the presence and localization of PCNA and bax positivity in the cells. Cells not treated with the primary antibody but only the secondary antibody and DAB were used as negative controls. The data from immunocytochemistry are reported as the means of values obtained in three separate experiments performed on separate days with different groups of granulosa cells, each obtained from at least five ovaries. In each chamber (one per group), 1000 cells were scored (i.e., each value representing the means of three replicates [3000 cells in total]). The percentage of DAB-HRP-stained cells to the total cell count was calculated. Significant differences between the groups were determined using the Shapiro–Wilk normality and Student's *t* test and two-way ANOVA, followed by Tukey's test with SigmaPlot 11.0 (Systat Software GmbH). Differences were compared for statistical significance at *p* levels less than .05 (*p* < .05).

2.4 | EIA/ELISA

The concentrations of progesterone, oxytocin, and PGF were determined in 25–100 μ l samples of the incubation medium by EIA, as described previously (Homanics & Silvia, 1988; Kotwica et al., 1994; Kotwica & Skarzynski, 1993). The cross-reactivity of antiserum against progesterone was <0.1% with 17 β -estradiol, dihydrotestosterone, testosterone, and 17 β -hydroxyprogesterone. Antiserum against oxytocin cross-reacted <0.01% with arginine vasopressin, lysine vasopressin, angiotensin, vasotocin, and somatostatin. 13,14-dihydro-15-keto-PGF2 α (PGFM) cross-reacted <0.1% with PGF2 α , PGE2, PGA2, and 6-keto-PGF1. The range of the standard

curve, the intra- and interassay coefficients of variation, and the relationship between the added and measured hormone concentrations, expressed as the coefficient of regression, were as follows: for progesterone, 0.37–25 ng/ml, 8.7%, 10.2%, and $r = .96$; for oxytocin, 3.9–1000 pg/ml, 9.8%, 10.8%, and $r = .94$; and for PGFM, 62.5–2000 pg/ml, 8.2%, 11.9%, and $r = .98$. The concentrations of testosterone and IGF-I were determined in 25 μ l aliquots of incubation medium by ELISA according to the manufacturer's instructions (LDN Immunoassays and Services). The cross-reactivity of antiserum against testosterone was $\leq 3.3\%$ with 11 β -hydroxytestosterone and 19-nortestosterone, $\leq 0.9\%$ with androstenedione, $\leq 0.8\%$ with 5 α -dihydrotestosterone, and $< 0.1\%$ with 17 α -methyltestosterone, epitestosterone, estradiol, progesterone, cortisol, estrone, and danazol. The maximal intra- and interassay coefficients of variation were 4.16% and 4.73%, respectively. The sensitivity of the assay was 0.083 ng/ml. Antiserum against IGF-I cross-reacted $\leq 3.3\%$ with insulin and 1.02% with IGF-II. The sensitivity of the assay was 9.75 ng/ml. Intra- and interassay coefficients of variation did not exceed 7.39% and 12.63%, respectively. The results from EIA/ELISA are reported as the means of values obtained in three separate experiments performed on separate days with different groups of granulosa cells, each obtained from at least five ovaries. Each experimental group was represented by four culture wells (i.e., each value represents the mean of four wells $\times$ 3 experiments = 12 replicates). The rates of substance secretion were calculated per 10^6 viable cells/day. Significant differences between the groups were determined using the Shapiro–Wilk normality and Student's t test and two-way ANOVA, followed by Tukey's test with SigmaPlot 11.0 (Systat Software GmbH). Differences were compared for statistical significance at p levels less than .05 ($p < .05$).

3 | RESULTS

3.1 | The effects of toluene, *Tribulus terrestris* extract, and their combination on bovine ovarian granulosa cell functions

The addition of toluene reduced the viability of bovine ovarian granulosa cells (at doses of 10 and 100 ng/ml, Figure 1a) and the accumulation of PCNA- (at doses of 10 and 100 ng/ml, Figure 1b)

and bax-positive (at doses of 1 and 10 ng/ml, Figure 1c) cells. Furthermore, toluene decreased the release of progesterone (at doses of 10 and 100 ng/ml, Figure 1d), testosterone (at doses of 10 and 100 ng/ml, Figure 1e), IGF-I (at a dose of 100 ng/ml, Figure 1f), oxytocin (at doses of 10 and 100 ng/ml, Figure 1g), and PGF (at doses of 1, 10, and 100 ng/ml, Figure 1h).

The addition of TT extract (at a dose of 10 μ g/ml) increased the percentage of PCNA-positive cells (Figure 1b) but did not alter cell viability (Figure 1a) or the percentage of bax-positive cells (Figure 1c). Furthermore, TT inhibited the output of progesterone (Figure 1d), oxytocin (Figure 1g), and PGF (Figure 1h) but did not alter testosterone (Figure 1e) or IGF-I (Figure 1f) release.

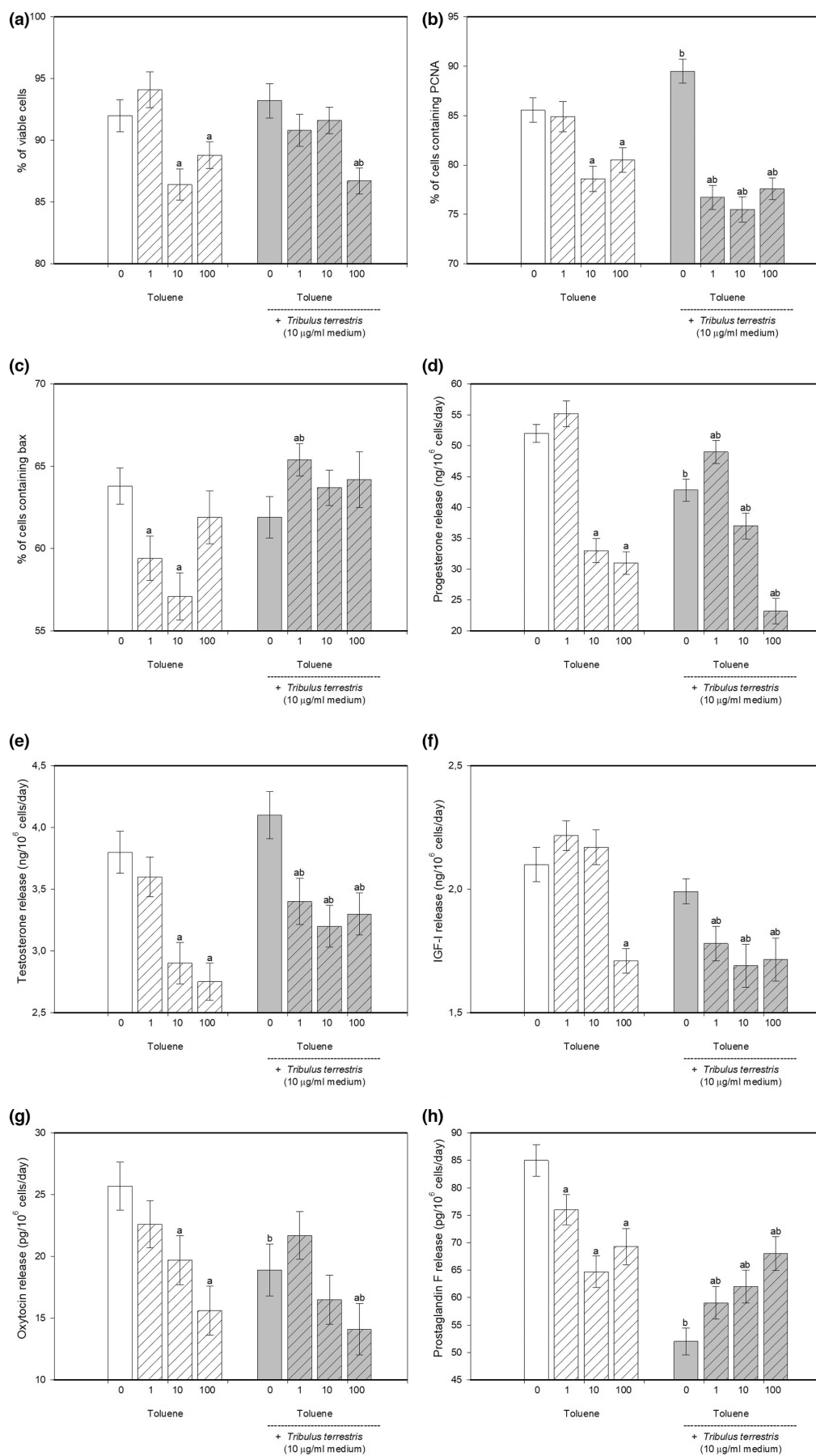
The presence of TT supported the toluene effect on cell viability (at a dose of 100 ng/ml, Figure 1a), the proportion of PCNA-positive cells (at doses 10 and 100 ng/ml, Figure 1b), the release of progesterone (at doses 10 and 100 ng/ml, but not at a dose 1 ng/ml, Figure 1d), testosterone (at doses 10 and 100 ng/ml, Figure 1e), IGF-I (at doses 1 and 10 ng/ml, Figure 1f), and PGF (at a dose of 1 ng/ml, Figure 1h). Furthermore, TT suppressed and even inverted the action of toluene on the accumulation of bax (at a dose of 1 ng/ml, Figure 1c) but did not change its influence on oxytocin output (Figure 1g).

Toluene induced the inhibitory effect of TT on cell viability (at a dose 100 ng/ml, Figure 1a), testosterone, and IGF-I release (at doses 1, 10, and 100 ng/ml, Figure 1e,f), supporting its inhibitory action on progesterone (at doses 1, 10, and 100 ng/ml, Figure 1d), oxytocin (at a dose 100 ng/ml, Figure 1g), and PGF (at doses 1, 10, and 100 ng/ml, Figure 1h) output. Furthermore, toluene suppressed and even inverted the stimulatory action of TT on the accumulation of PCNA (at doses 1, 10, and 100 ng/ml, Figure 1b), and induced the stimulatory effect of TT on the proportion of bax-positive cells (at a dose of 1 10 ng/ml, Figure 1c).

3.2 | The effects of toluene, *Tribulus terrestris* extract, and their combination on equine ovarian granulosa cell functions

The addition of toluene to the culture medium resulted in a decrease in the cell viability of equine ovarian granulosa cells (at doses of 10 and 100 ng/ml, Figure 2a), the percentage of proliferation

FIGURE 1 Effects of the addition of toluene (0, 1, 10, 100 ng/ml) alone, *Tribulus terrestris* extract (TT) (10 μ g/ml), or toluene (1, 10, 100 ng/ml) in combination with TT (10 μ g/ml) on cell viability (trypan blue exclusion assays) (a), proliferation (accumulation of PCNA, quantitative immunocytochemistry) (b), apoptosis (accumulation of bax, quantitative immunocytochemistry) (c), and the release of progesterone (EIA) (d) in cultured bovine ovarian granulosa cells. The results are expressed as mean $\pm$ SEM. The results show (a) the effect of toluene, with a significant ($p < .05$) difference between the cells cultured without or with toluene, (b) the effect of TT extract, with a significant ($p < .05$) difference between the cells cultured without or with TT extract, and (ab) the effect of toluene with simultaneous TT presence, with a significant ($p < .05$) difference between the cells cultured with toluene with simultaneous TT extract presence. Effects of the addition of toluene (0, 1, 10, 100 ng/ml) alone, *Tribulus terrestris* extract (TT) (10 μ g/ml), or toluene (1, 10, 100 ng/ml) in combination with TT (10 μ g/ml) on the release of testosterone (ELISA) (e), IGF-I (ELISA) (f), oxytocin (EIA) (g) and prostaglandin F (EIA) (h) in cultured bovine ovarian granulosa cells. The results are expressed as mean $\pm$ SEM. The results show (a) the effect of toluene, with a significant ($p < .05$) difference between the cells cultured without or with toluene, (b) the effect of TT extract, with a significant ($p < .05$) difference between the cells cultured without or with TT extract, and (ab) the effect of toluene with simultaneous TT presence, with a significant ($p < .05$) difference between the cells cultured with toluene with simultaneous TT extract presence



marker (PCNA) (at doses of 1, 10, and 100 ng/ml, Figure 2b), and the marker of apoptosis (bax) (at doses of 10 and 100 ng/ml, Figure 2c). Furthermore, toluene inhibited the release of progesterone (at doses of 1, 10, and 100 ng/ml, Figure 2d), testosterone (at a dose of 100 ng/ml, Figure 2e), IGF-I (at doses of 1, 10, and 100 ng/ml, Figure 2f), oxytocin (at a dose of 100 ng/ml, Figure 2g), and PGF (at doses of 1, 10, and 100 ng/ml, Figure 2h).

The addition of TT extract (at a dose of 10 µg/ml) did not change the cell viability (Figure 2a) or the proportion of apoptotic (bax-positive) cells (Figure 2c) and promoted the proportion of granulosa cells containing PCNA (Figure 2b). In addition, TT reduced progesterone (Figure 2d), testosterone (Figure 2e), IGF-I (Figure 2f), oxytocin (Figure 2g), and PGF (Figure 2h) output.

The presence of TT supported the toluene effect on cell viability (at doses of 10 and 100 ng/ml, Figure 2a), the release of progesterone (at doses of 1, 10, and 100 ng/ml, Figure 2d), testosterone (at doses of 1 and 10 ng/ml, Figure 2e), IGF-I (at doses of 1 and 10 ng/ml, Figure 2f), oxytocin (at doses of 1, 10, and 100 ng/ml, Figure 2g), and PGF (at doses of 1 and 10 ng/ml, Figure 2h). Furthermore, TT suppressed the influence of toluene on the accumulation of PCNA (at doses of 1, 10, and 100 ng/ml, Figure 2b) and inverted its action on the percentage of bax-positive cells (at doses of 10 and 100 ng/ml, Figure 2c).

Toluene induced the inhibitory effect of TT on cell viability (at doses of 1, 10, and 100 ng/ml, Figure 2a), supporting its inhibitory action on progesterone (at a dose of 10 ng/ml, Figure 2d), testosterone (at a dose of 100 ng/ml, Figure 2e), IGF-I (at doses of 1, 10, and 100 ng/ml, Figure 2f), and PGF (at a dose of 10 ng/ml, Figure 2h) release. Furthermore, toluene suppressed the stimulatory influence of TT on the proportion of PCNA-positive cells (at a dose of 100 ng/ml, Figure 2b) and induced its stimulatory effect on the accumulation of bax (at a dose of 1 ng/ml, Figure 2c) but did not change its action on oxytocin output (Figure 2g).

4 | DISCUSSION

The presence of proliferation- and apoptosis-related molecules, as well as the ability of cells to exclude trypan blue and to secrete steroid and peptide hormones and PGF into the culture medium,

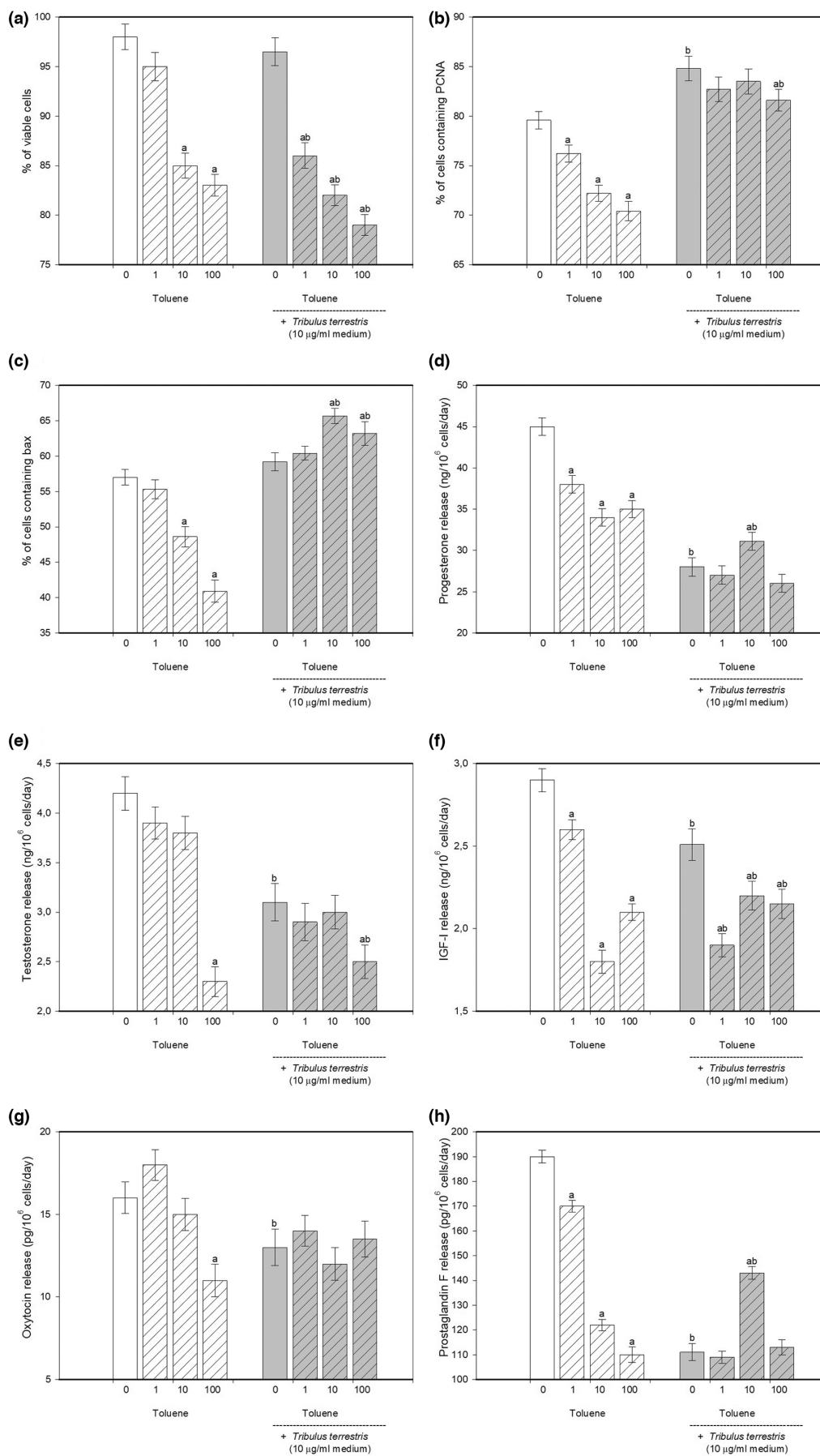
indicates that the tested equine and bovine ovarian cells were viable and suitable for analysis. Furthermore, the observed changes demonstrate the direct influence of toluene and TT on cell viability, proliferation, apoptosis, and the release of ovarian hormones. These values are addressed as both markers and regulators of female reproductive processes and fecundity. Furthermore, an increase in progesterone release by cultured granulosa cells indicates that these cells were subjected to luteinization in the culture (Sirotkin, 2014).

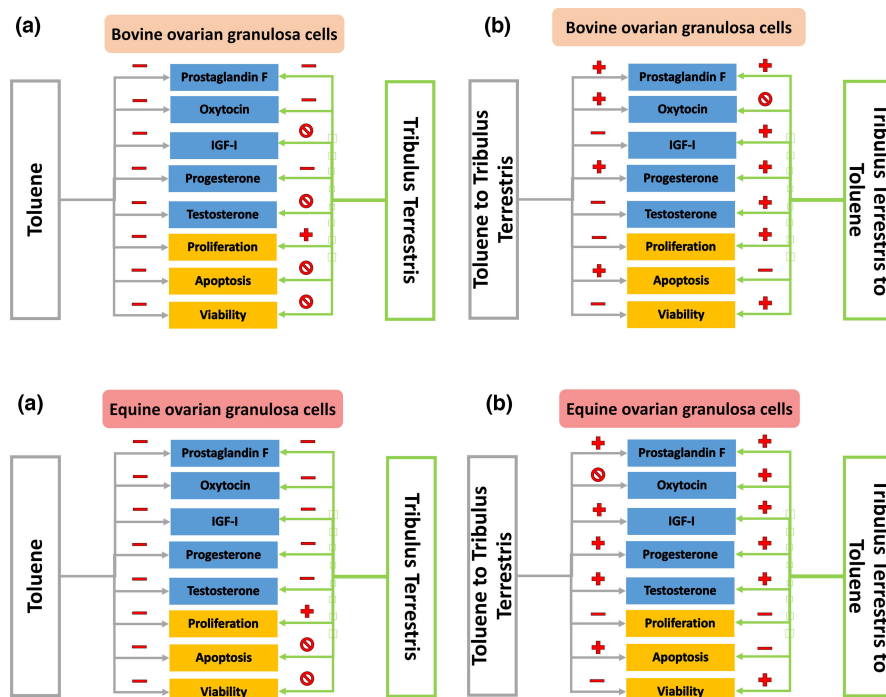
4.1 | The effect of toluene on ovarian cell functions

In our experiments, the addition of toluene inhibited cell viability, proliferation (accumulation of PCNA-positive cells), and accumulation of bax-positive cells (marker and promoter of intrinsic apoptosis) (Peña-Blanco & García-Sáez, 2018), as well as progesterone, testosterone, IGF-I, oxytocin, and PGF release by both bovine and equine ovarian cells (Scheme 1a and 2a). This indicates that this oil-related pollutant can directly influence ovarian cell viability, proliferation, apoptosis, and secretory activity in both of these domestic animals. These observations confirmed our previous findings on the inhibitory influence of toluene on cell viability, proliferation, and apoptosis in porcine (Sirotkin, Kadasi, et al., 2020; Sirotkin, Záhoranská, et al., 2021), equine (Tarko et al., 2020), and bovine (Sirotkin, Makarevich, et al., 2020) ovarian cells. The toluene-induced decrease in proliferation rates and inhibition of apoptosis are not characteristic of cells undergoing malignant transformation. Therefore, our observations do not indicate the potential carcinogenic action of toluene on ovarian cells. This conclusion is in line with the results of previous studies on toluene (Report on Carcinogens, 2016; Soffritti et al., 2006), which did not reveal a carcinogenic effect of this oil-related pollutant. Based on these observations, the National Toxicology Program has not classified toluene as a carcinogen (Report on Carcinogens, 2016).

The PCNA protein is considered a promoter of DNA polymerase action and regulator of the S-phase of the cell cycle (Shiomi & Nishitani, 2017). Therefore, toluene may suppress the S-phase of mitosis via the reduction of PCNA accumulation. Bax is a well-known regulator and marker of cytoplasmic apoptosis (Peña-Blanco & García-Sáez, 2018); therefore, it might be hypothesized that

FIGURE 2 Effects of the addition of toluene (0, 1, 10, 100 ng/ml) alone, *Tribulus terrestris* extract (TT) (10 µg/ml), or toluene (1, 10, 100 ng/ml) in combination with TT (10 µg/ml) on cell viability (trypan blue exclusion assays) (a), proliferation (accumulation of PCNA, quantitative immunocytochemistry) (b), apoptosis (accumulation of bax, quantitative immunocytochemistry) (c), and the release of progesterone (EIA) (d) in cultured equine ovarian granulosa cells. The results are expressed as mean ± SEM. The results show (a) the effect of toluene, with a significant ($p < .05$) difference between the cells cultured without and with toluene, (b) the effect of TT extract, with a significant ($p < .05$) difference between the cells cultured without and with TT extract, and (ab) the effect of toluene with simultaneous TT presence, with a significant ($p < .05$) difference between the cells cultured with toluene with simultaneous TT extract presence. Effects of the addition of toluene (0, 1, 10, 100 ng/ml) alone, *Tribulus terrestris* extract (TT) (10 µg/ml), or toluene (1, 10, 100 ng/ml) in combination with TT (10 µg/ml) on the release of testosterone (ELISA) (e), IGF-I (ELISA) (f), oxytocin (EIA) (g) and prostaglandin F (EIA) (h) in cultured equine ovarian granulosa cells. The results are expressed as mean ± SEM. The results show (a) the effect of toluene, with a significant ($p < .05$) difference between the cells cultured without and with toluene, (b) the effect of TT extract, with a significant ($p < .05$) difference between the cells cultured without and with TT extract, and (ab) the effect of toluene with simultaneous TT presence, with a significant ($p < .05$) difference between the cells cultured with toluene with simultaneous TT extract presence





SCHEME 1 Character of effect of toluene and *Tribulus terrestris* (TT) given alone (a) on cultured bovine ovarian granulosa cell functions, as well as the ability of TT to modify toluene effects and vice versa (b)

SCHEME 2 Character of effect of toluene and *Tribulus terrestris* (TT) given alone (a) on cultured equine ovarian granulosa cell functions, as well as the ability of TT to modify toluene effects and vice versa (b)

toluene reduces this type of apoptosis. The equilibrium between proliferation and apoptosis/death defines cell viability and determines whether a follicle will continue to develop or be diverted into atretic pathways (Robker & Richards, 1998; Voronina et al., 2007). Our findings demonstrated that toluene is able to inhibit ovarian follicular cell proliferation and apoptosis, which is an indication of decreased ovarian cell turnover. Such inhibition of cell turnover can explain the ability of this oil-related pollutant to reduce cell viability. Notably, the evidence that toluene can support bovine and equine ovarian follicular atresia and suppress ovarian folliculogenesis, and therefore fecundity, should not be excluded. In addition, our observation confirmed previous studies concerning the inhibitory action of toluene on the release of progesterone, IGF-I, and PGF by cultured equine (Tarko et al., 2020) and its adverse effects on progesterone, estradiol, and oxytocin output by porcine ovarian cells (Sirotkin, Kadasi, et al., 2020; Sirotkin, Záhoranská, et al., 2021). Furthermore, our observations are potentially the first to demonstrate the direct toluene effect on the release of testosterone in horses and on the release of testosterone, oxytocin, and PGF in cows.

The functional interrelationships between different steroid hormones, IGF-I, oxytocin, and PGF affected by toluene require further clarification. Since ovarian hormones and growth factors, including progesterone, testosterone, oxytocin, PGF, and IGF-I, are key regulators of ovarian cell viability, proliferation, apoptosis and the resulting ovarian folliculogenesis, oogenesis, and fecundity (Sirotkin, 2014), it is possible that toluene may affect these processes by up- or down-regulating the release of these substances. For example, the effect of toluene on some hormones could be primary but on other hormones secondary through changes in hormone precursors. Nevertheless, it might be hypothesized that the decrease in testosterone release

could be due to a decrease in the production of its precursor progesterone (Das & Kumar, 2018). In addition, these results suggest that toluene can affect bovine and equine ovarian steroidogenesis and reproduction via other non-steroid hormones – IGF-I, oxytocin, and PGF. In particular, both oxytocin and PGF are regulators of steroid hormones and IGF-I secretion (Sirotkin, 2014). Therefore, a decrease in steroids and IGF-I release could be due to a reduction in oxytocin and PGF output. Furthermore, toluene can be a regulator of ovarian functions via changes in ovarian cell proliferation and apoptosis, which, in turn, could be induced by changes in ovarian steroidogenesis, IGF-I, oxytocin, and PGF release. Our observations may indicate toluene involvement in suppressing ovarian follicle selection and their further development, which are characterized by decreased ovarian cell proliferation and apoptosis (Sirotkin, 2014). Furthermore, toluene could be involved in the reduction of ovarian cell luteinization, which is defined by a decrease in ovarian cell proliferation, progesterone, and PGF release (Sirotkin, 2014). These assumptions are supported by the *in vivo* study of Tap et al. (1996), who reported degenerative changes in antral follicles of rats attributable to the inhalation of toluene. It should be noted that in our study, toluene treatment reduced the release of testosterone, which suggests that toluene can inhibit testosterone-related ovarian atresia associated with apoptosis (Sirotkin, 2014). These observations correspond to previous reports that continuous exposure to toluene results in degenerative histomorphological changes in mouse ovarian tissue, which might be due to disruption of endocrine function (Kareem et al., 2014). These oil-related, pollutant-induced changes indicate possible cellular and endocrine mechanisms of harmful toluene influences on female reproduction reported previously (see Introduction) and might participate in reproductive toxicity and infertility. Nevertheless, a better understanding of the characteristics

and mechanisms of toluene action on reproduction and fecundity requires further in vitro and in vivo studies.

4.2 | The effect of *Tribulus terrestris* on ovarian cell functions

In the current study, we evaluated the effects of TT extract on bovine and equine reproduction. In both farm animals, TT promoted proliferation and reduced the release of progesterone, oxytocin, and PGF but did not change cell viability or apoptosis. In bovine ovarian cells, TT did not affect testosterone or IGF-I output; Yet, in equine cells, it reduced both of these ovarian hormones (Scheme 1a and 2a). Our findings contradict the previous results of Sirotkin, Alexa, and Harrath (2020), who reported the ability of TT to increase the accumulation of bax in porcine ovarian cells. The difference in the character of TT action on ovarian cell apoptosis observed in different experiments could be explained by the difference in species (cows and horses versus pigs by Sirotkin, Alexa, & Harrath, 2020).

Our finding suggests that the influence of TT on the fundamental processes previously observed in ovarian cells (Abadjieva & Kistanova, 2016; Arentz et al., 2014; Dehghan et al., 2012; Sandeep et al., 2015) could be due to the direct effects of its constituents on ovarian cells. Understanding the mechanisms of TT influence on ovarian cells requires further study, but the available information enables us to suggest some mediators of this TT action. For example, TT extract contains various medicinally important phytochemicals, such as steroidal saponins, phytosterols, and flavonoids (Chhatre et al., 2014). It is possible that the stimulatory action of TT on cell proliferation arises from the presence of saponins, which are able to enhance cell membrane permeability and, therefore, the transport of nutrients and other molecules into cells (Johnson et al., 1986). In addition, the antioxidant and phytoestrogenic properties of TT saponins, including diosgenin and flavonoids, including rutin (Chhatre et al., 2014) could, in turn, promote cell proliferation and prevent DNA fragmentation, resulting in apoptosis. Such an activity has been observed in previous experiments: diosgenin and rutin were able to increase proliferation of cultured porcine ovarian granulosa cells (Sirotkin, Alexa, et al., 2019; Sirotkin, Záhoranská, et al., 2020). Furthermore, in our study, TT promoted ovarian cell proliferation and did not affect apoptosis, which, in turn, can result in the promotion of ovarian follicular growth and development (Sirotkin, 2014). Therefore, one could expect the promotion of proliferation, which is not associated with changes in apoptosis, to be associated with the improvement of cell viability. Yet, in our study, TT did not influence cell viability. We hypothesize that this lack of effect on cell viability could be due to neutralization of increased proliferation, not by cytoplasmic apoptosis (which is characterized by increased accumulation of bax [Peña-Blanco & García-Sáez, 2018] measured in our experiments), but by nuclear apoptosis/DNA fragmentation, or autophagy. The markers of these processes were not determined in our study; nevertheless, it is possible that they could be capable of inducing

cell death without bax commitment (Kitazumi & Tsukahara, 2011; Nikolettópoulou et al., 2013).

Tribulus terrestris L. (TT) extract is also rich in flavonoid such as quercetin (Chhatre et al., 2014). It is likely that the inhibitory effects of TT on progesterone and testosterone output observed in our experiments resulted from the presence of quercetin and diosgenin, given that previous studies (Sirotkin, Alexa, et al., 2019; Sirotkin, Alexa, et al., 2021; Sirotkin, Hrabovszká, et al., 2019; Sirotkin, Štochmařová, et al., 2019; Tarko et al., 2018) have shown inhibitory action of this flavonoid on these steroids. Since progesterone is a known inhibitor of granulosa cell mitosis and suppressor of their apoptosis, and an inhibitor of ovarian follicular development, ovulation, and promoter of follicular cell luteinization (Sirotkin, 2014), it is likely that TT influences these processes via inhibition of progesterone release. Our observation is in line with the results of a previous study by Bertozzo and Moreira (2019), which demonstrated the inhibitory effect of TT on blood testosterone levels in mice. We could not exclude the possibility that TT also alters ovarian functions through the inhibition of testosterone, which can promote ovarian cell atresia linked with apoptosis (Sirotkin, 2014). Alternatively, it has been proposed that TT active components might be enzymatically converted from progesterone into testosterone, while testosterone is aromatized into estradiol (Das & Kumar, 2018; Sirotkin, Alexa, & Harrath, 2020). Therefore, a decrease in testosterone could occur due to the inhibition of progesterone or due to its increased conversion to estradiol.

The TT-induced changes could be due to changes in the release of the non-steroid hormones IGF-I, oxytocin, and PGF, which are potent stimulators of cellular proliferation, differentiation, and development, steroidogenesis, and inhibitors of apoptosis during follicular development (Sirotkin, 2014). It could be hypothesized that the TT-induced decline in both progesterone and testosterone release could be due to a reduction in the production of their regulators. On the other hand, the lack of influence of TT on cell cytoplasmic apoptosis in our experiments suggests that changes in IGF-I, oxytocin, and PGF release are unable to affect certain IGF-I-, oxytocin-, and PGF-dependent processes.

In addition, the effective doses of TT were different in different species, indicating that cells isolated from ovaries of various species had different responses to TT. Understanding the causes and biological significance of such inter-species differences in TT action on ovarian hormones requires additional study. It might be hypothesized that ovarian cells of different species have different numbers of estrogen receptors that are involved in the mediation of the phytoestrogenic properties of TT (Chhatre et al., 2014). Furthermore, in some species, there may exist negative feedback mechanisms that inhibit the cell response to steroids or other hormones if this regulator is present in higher amounts than are optimal for cellular responses (Sirotkin, 2014).

In conclusion, our observations are the first demonstration of the direct influence of TT on healthy bovine and equine ovaries. The suppressive effects of TT basic ovarian cell functions suggest a potential negative influence of TT on livestock reproduction. If

similar conclusions would be reached in in vivo experiments in both farm animals and humans, the anti-reproductive action of TT should be taken into account when considering the consumption of TT-containing foods or drugs. In addition, this action of TT suggests its potential usefulness for the regulation and synchronization of ovarian cycles and treatment of some reproductive and steroid-related disorders, such as ovarian hyperstimulation syndrome. The applicability of TT as an alternative form of contraception could also not be excluded.

4.3 | Interrelationships between toluene and *Tribulus terrestris* in the control of ovarian cell functions

Toluene induced the inhibitory effect of TT on cell viability, supported its inhibitory action on the release of progesterone and PGF, suppressed its stimulatory influence on proliferation, and induced its stimulatory effect on apoptosis in both animals. In bovine cells, toluene induced the inhibitory action of TT on testosterone and IGF-I and supported its effect on oxytocin output. In equine cells, toluene supported the inhibitory effect of TT on testosterone and IGF-I but did not change its action on oxytocin output (Scheme 1b and 2b). These observations suggest that toluene can influence not only basic bovine and equine ovarian cell functions but also their response to food/medicinal plant TT. Toluene and TT looked as synergists in their suppressive actions on ovarian cells.

Some studies demonstrated that the adverse effects of environmental pollutants such as toluene on the female reproductive system could be prevented or neutralized by the application of medicinal plant extract (Tarko et al., 2020) or plant flavonoids (Sirotkin, Záhoranská, et al., 2021). Previous animal studies suggest that TT, due to its antioxidative properties, may provide effective protection against the toxic effects of cadmium (Lakshmi et al., 2012), lead acetate (Eagappan et al., 2015), nifedipine (Aslam, 2018), and cisplatin (Keshtmand et al., 2014; Raoofi et al., 2015). In our experiments, TT supported the toluene action on cell viability, the release of progesterone, testosterone, IGF-I, and PGF output. Furthermore, TT was able to be suppressed and even inverted the action of toluene on apoptosis in both species. There were substantial species-dependent differences in the ability of TT to modify the ovarian response to toluene: in cows, TT supported the toluene effect on proliferation and did not alter oxytocin output, while in horses TT promoted its effect on oxytocin release and suppressed its action on proliferation (Scheme 1b and 2b). Our observations represent the first evidence that TT can be a plant-based protector against the adverse action of toluene on healthy ovaries.

Understanding the physiological importance of these parameters and the protective mechanism of TT against oil-related contaminants requires further exploration. Environmental pollutants, including toluene, can promote the generation of reactive oxygen species (ROS) (Ashraf et al., 2011). ROS, when present in

high concentrations, can overwhelm cellular antioxidants and lead to oxidative stress (Schieber & Chandel, 2014). Cells exposed to oxidative stress undergo extensive cellular damage due to DNA breakage, lipid (membrane) oxidation, and extensive protein damage (Valko et al., 2004). *Tribulus terrestris* L. (TT) contains antioxidants (Chhatre et al., 2014), which are capable of preventing the potentially damaging effects of toluene-induced ROS, by blocking pollutant-induced ROS. In addition, the protective action of TT could be attributed to the presence of steroidal saponins and phytoestrogens (phytosterols) (Chhatre et al., 2014), which can interfere with granulosa cell steroidogenic activity (Sirotkin, 2014). This suggestion was confirmed by TT's ability to affect and modify the action of toluene on progesterone and testosterone release. Another mechanism of the protective effects of TT phytoestrogens is their ability to bind and alter the activity of steroid hormone receptors (Sirotkin, 2014). All hypotheses concerning the mechanisms of TT action on toluene effects require experimental validation.

Taken together, our experiments demonstrated that the negative effects of toluene on bovine and equine ovarian cell viability, proliferation, progesterone, testosterone, IGF-I, and oxytocin release, processes that are essential for the control of reproduction (Sirotkin, 2014), can be prevented by TT. Therefore, their impairment by toluene can suppress reproductive processes, while TT could be a natural protector against the negative effects of toluene and thus maintain and improve the reproductive functions of livestock even in conditions of environmental contamination by oil-related pollutants. The details of toluene and TT action, their functional interrelationship, and possibilities for their application are worthy of future study. Nevertheless, the present observations (1) confirm the toxic influence of toluene on ovarian cells, (2) demonstrate the direct effect of TT on these cells, (3) indicate the ability of toluene to promote and induce the main effects of TT, (4) demonstrate the ability of TT to prevent the action of toluene on cell apoptosis and proliferation, and promote its effect on cell viability, steroids, IGF-I, oxytocin, and PGF, and (5) show the similar response of both bovine and equine ovarian cells to toluene and their differing responses to TT and its combination with toluene. Understanding these mechanisms may be helpful in the regulation of animal and human reproduction and in the prevention/treatment of the negative effects of environmental contaminants on reproductive processes.

AUTHOR CONTRIBUTIONS

Z. Fabová and A. Tarko curated, visualized and investigated the data; wrote the original draft; and reviewed and edited the manuscript. A.H. Harrath visualized, investigated and validated the data; assisted choosing the software; and reviewed and edited the manuscript. S. Alwasel and J. Kotwica conceptualized the study; designed the methodology of the study; assisted choosing the software; and validated the data. A.V. Sirotkin conceptualized the study; designed the methodology of the study; assisted choosing the software; supervised the study; and reviewed and edited the manuscript.

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CONFLICT OF INTEREST

None of the authors have any conflict of interest to declare.

DATA AVAILABILITY

The authors confirm that the data supporting the findings of this study are available within the article and its supplementary materials.

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